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Difference between Soil fertility Vs Soil productivity

Soil fertility	Soil productivity
• It is the ability of a soil to supply all the essential plant nutrients in readily available forms to the crop plants • It is considered as an index of available nutrients. • It is one of the factor for crop production and other factors are moisture, topography, water table, tillage practice, varieties, organic matter content of soil etc.	• It is the capacity of the soil to produce crops per unit area of the field. • It is the broader term used to indicate the yield of crops. $Y = f(x_1, x_2, \dots, x_n)$
• It can be analyzed in the lab.	• It is the interaction of all the factors that determine the magnitude of yield.
• It is the potential status of soil to produce the crops.	• It can be assessed in the field in a particular climate and system of management. • It is the resultant of various factors influencing soil management.

- **Ions-** Which carry charge (Ca^{++} , Mg^{++} , NO_3^- , H_2PO_4 , HPO_4)
 - Which carry **positive charge**- Cations (Ca^{++} , Mg^{++} , Na^+ etc.)
 - Which carry **negative charge**- Anions (NO_3^-), H_2PO_4 , HPO_4 etc.
- **Minerals-** are those which derive from rocks or soils.
 - CHO are not minerals because **do not absorb from soil**.
- **Nutrients** are those which participate in plant growth.
- **Nutrition-** it is a process which follows three steps absorption, translocation and assimilation.
- If nutrients is absorb and translocate but not assimilate **not called nutrition**.
- **Assimilation** means nutrients convert in cell structure components like N in protein (protein is assimilation end).

Criteria of essential nutrients

Criteria of essential nutrients proposed by **Arnon and Stout in 1939** and further **refined by Arnon 1954**.

- **Three Criteria for Essentiality of Nutrients:**

1. A plant **cannot complete its life-cycle** in the absence of the element
2. The action of the element must be specific, with no other element being able to completely substitute for it (**can not replace**).
3. The element must be shown to be **directly involved in the nutrition or metabolism of the plant**. It must be a constituent of a metabolic pathway or at least be required for the activity of an essential enzyme.

Classification of nutrients based on the requirement of nutrients

1. Macronutrients

- Plants need large amounts of micronutrients relative to other essential nutrients.
- **Three structural are:** Carbon (C), Hydrogen (H), Oxygen (O)
- **Six macronutrients are from soil:** Nitrogen (N), Phosphorus (P), Potassium (K), Magnesium (Mg), Calcium (Ca), Sulfur (S)

2. Micronutrients

1. Compared with macronutrients, the concentrations of the micronutrients in plants may be very small. **This does not mean they are not important.** Deficiencies or excesses of the micronutrients can cause yield loss just as macronutrient deficiencies or toxicities. Micronutrients are often considered enzyme nutrients:
 - All micronutrients are from soil
 - **Six micronutrients are cations:** Copper (Cu), Manganese (Mn), Iron (Fe), Zinc (Zn), Nickel (Ni), Cobalt (Co)
 - **Three micronutrients are anions:** Chloride (Cl), Boron (B), Molybdenum (Mo).

Functional nutrient

- According to this criterion, **sodium** is considered as **nonessential**.
- However, sodium is known to **increase yield** of **several crops such as sugar beet, turnip and celery**.

Therefore, a farmer considers sodium as an essential element.

- **Nicholas (1961) proposed the term 'functional nutrient' for any mineral element**
- **that functions in plant metabolism whether or not its action is specific.**
- With this criterion, **sodium, cobalt, vanadium, nickel and silicon** are also considered as functional nutrients in addition to 17 essential elements.

Relative and Average Plant Nutrient Concentrations

Nutrients	Average concentration in plant tissue	Function in plant	Nutrient category
N	1.5%	Proteins, amino acids	Primary Macronutrients
P	0.2%	Nucleic acids, ATP	
K	1.0%	Catalyst, ion transport	
Ca	0.5%	Cell wall component	Secondary
Mg	0.2%	Part of chlorophyll	
S	0.1%	Amino acids	
Fe	100 mg/kg	Chlorophyll synthesis	Micronutrients
Cu	6 mg/kg	Component of enzymes	
Mn	20 mg/kg	Activates enzymes	
Zn	20 mg/kg	Activates enzymes	
B	20 mg/kg	Cell wall component	
Mo	0.1 mg/kg	Involve in N fixation	
Cl	100 mg/kg	Photosynthesis reactions	

*mg/kg is also known as parts per millions (ppm)

(ii) According to mobility:

(a) In soil:

1. Mobile: NO_3^- , SO_4^{2-} , BO_3^{3-} , Cl^- and Mn^{2+}

2. Less mobile: NH_4^{2+} , K^+ , Ca^{2+} , Mg^{2+} and Cu^{2+}

3. Immobile: H_2PO_4^- , HPO_4^{2-} and Zn^{2+}

- **Nutrients mobility in soil depends upon**
 - Ion charge types
 - Reactivity of ion in soil
- Mobile nutrients generally anions due to solubility.
- Less mobile nutrients include cations due to adsorb in clay.
- Immobile or poor- Due to high activity of these ions like H_2PO_4^- and Zn^{2+}
- The mobility of a nutrient in the soil determines how much can be lost due to leaching or runoff.

Mobility of nutrients in plants

(b) In plant:

- 1. Highly mobile:** N, P and K
- 2. Moderately mobile:** Zn
- 3. Less mobile:** S, Fe, Mn, Cu, Mo and Cl
- 4. Immobile:** Ca and B

- **Mobility** of nutrients in plants means ability of nutrients to **translocation and redistribution**.
- The mobility of a nutrient in the plant determines **where deficiency symptoms show up**.
 - **Nutrients that are mobile in the plant** will move to new growth areas, so the deficiency symptoms will first show up in **older leaves**.
 - Nutrients that are **not mobile in the plant** will not move to new growth areas, so deficiency symptoms will **first show up in the new growth**.

- For nutrients like **nitrogen, phosphorus, and potassium**, which are mobile in the plant, deficiency symptoms will appear in **older leaves**.
- As new leaves develop, they will take the nutrients from the old leaves and use them to grow.
- The old leaves are then left without enough nutrients, and display the symptoms.
- The opposite is true of **immobile nutrients like calcium and boron** the new leaves will have symptoms first because they cannot take nutrients from the old leaves, and there is not enough in the soil for their needs.

Essentiality of nutrients discovered by scientists

Nutrients	Essentiality discovered authors
C	Priestley <i>et al.</i>
N	Theodore de Saussure
K, Ca, Mg & P	C. Sprengel
S	Sachs and Knop
Cl	T.C. Broyer, A.B. Carlton, CM. Johnson and P.R. Stout
Fe	E Gris
B	K. Warington
Mn	J.S. McHargue
Zn	A.L. Sommer and CP. Lipman
Cu	A.L. Sommer, CP. Lipman and G. McKinney
Mo	D.I. Arnon and P.R. Stout
Ni	P.H. Brown, R.M. Welch and F.F. Cary